Tour de Maths 2022
Leg 5
09 June

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Beaulieu College

ALIS VOLA PROPRIIS

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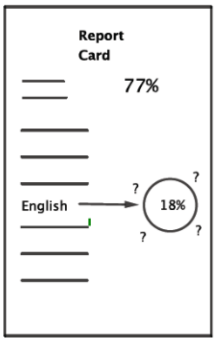
Question 1

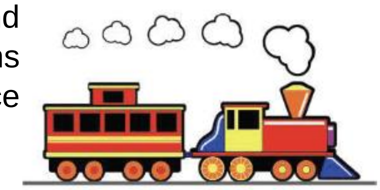
Mildred has seven grades on her report card. The overall average of these seven grades is 77%



After looking more closely at her report card, Mildred discovered that her English grade was incorrectly recorded as 18% instead of her actual grade of 81%. Determine Mildred's correct report card average.

Question 2

At noon, two trains are 1000 km apart. The first train is north of the second train and is traveling south. The second train is traveling east. Both trains travel at the same average speed, 100 km/h. At what time is the total distance travelled by the two trains equal to the distance between the trains?



Question 3

Alex can choose four different numbers *w, x, y* and *z* from the set $\{-1, -2, -3, -4, -5\}$.

What is the largest possible value of: $w^x + y^z$

Question 4

Greta currently works 45 hours per week and earns a weekly salary of R729. She will soon be starting a new job where her salary will be increased by 10% and her hours reduced by 10%. How much more will she be earning per hour at her new job?

Question 5

A train is 1000 metres long. It is travelling at a constant speed, and approaches a tunnel that is 3000 metres long. From the time that the last car on the train has completely entered the tunnel until the time when the front of the train emerges from the other end, 30 seconds pass.

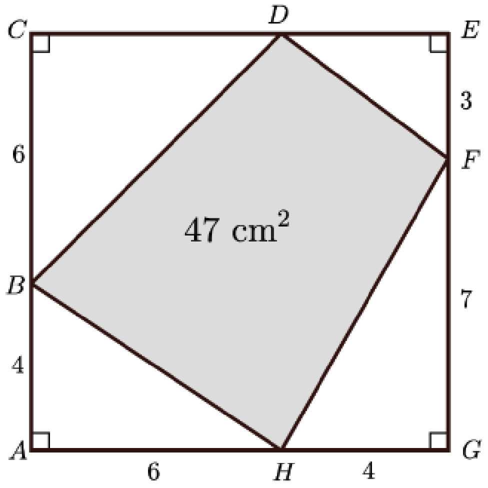
Determine the speed of the train, in kilometres per hour.

Question 6

Quadrilateral BDFH is constructed so that each vertex is on a different side of square AC EG. Vertex B is on side AC so that AB = 4 cm and BC = 6 cm.

Vertex F is on EG so that EF = 3 cm and FG = 7 cm. Vertex H is on GA SO that GH = 4 cm and HA= 6 cm. The area of quadrilateral BDFH is 47 cm2.

The fourth vertex of quadrilateral BDFH, labelled D, is located on side CE so that the lengths of CD and DE are both positive integers. Determine the length of CD.



Question 7

Bryn's Grandpa is always creating math problems for her to solve. In one problem, he gave her a store receipt that listed 72 identical items, each the same price, for a total cost of

R★67.9★

Bryn's Grandpa covered the first and last digits of the total price on the receipt with stars. Determine the sum of the digits that Bryn's Grandpa covered.

Question 8

In a magic square, the numbers in each row, the numbers in each column, and the numbers on each diagonal have the same sum. In the magic square shown, the sum a + b + c equals?

a	13	b
19	c	11
12	d	16

Question 9

When the expression $10^{2021} - 2021$ is evaluated, the result is a very large number.

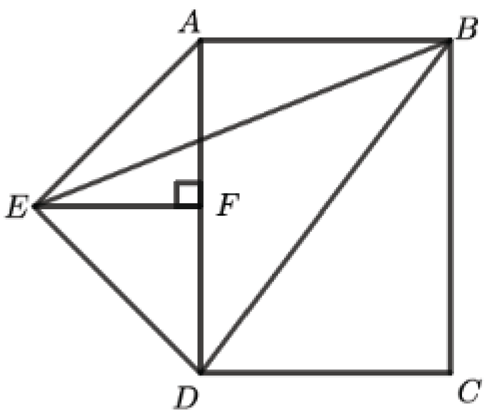
You probably do not have enough time to perform the calculation. So, in an effort to save you some time and paper, instead of evaluating the expression $10^{2021} - 2021$, determine the sum of the digits in the difference.

$$10^{2021} - 2021$$

Question 10

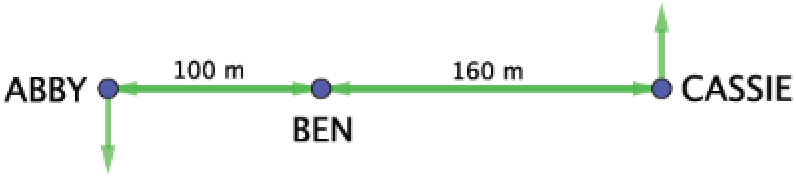
In the diagram below, ABCD is a rectangle. Point E is outside the rectangle so that ΔAED is an isosceles right-angled triangle with hypotenuse AD. Point F is the midpoint of AD, and EF is perpendicular to AD.

If BC = 4 and AB = 3, determine the area of ΔEBD.



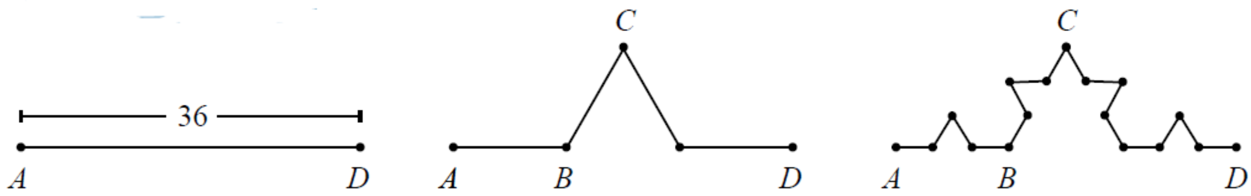
Question 11

At noon three students, Abby, Ben, and Cassie, are standing so that Abby is 100 m west of Ben and Cassie is 160 m east of Ben. While Ben stays in his initial position, Abby begins walking south at a constant rate of 20 meters per minute and Cassie begins walking north at a constant rate of 41 meters per minute. In how many minutes will the distance between Cassie and Ben be the twice the distance between Abby and Ben?



Question 12

Lin starts with a line segment that has length 36 and adds a bump to it. She then adds bumps to each line segment of that path. The resulting figure is shown below. All segments in each figure are of equal length. What is the total length of the figure 3?



Question 13

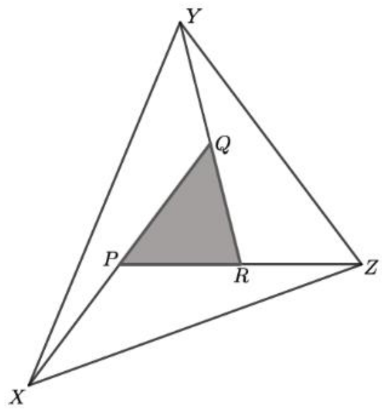
The sides of ΔPQR are extended to create ΔXYZ as follows:

side RQ is extended to Y so that RQ = QY,

side QP is extended to X so that QP = PX, and

side PR is extended to Z so that PR= RZ.

If the area of ΔXYZ = 1197, determine the area of ΔPQR.



Question 14

The number A8 is a two-digit nurnber with tens digit A and units (ones) digit 8. Similarly, 3 B is a two-digit number with tens digit 3 and units digit B. When A8 is multiplied by 3B, the result is the four-digit number C730. That is,

$$\begin{array}{r} A8 \\ \times 3B \\ \hline C730 \end{array}$$

If A, B, and C are each different digits from 0 to 9, determine the value of A+B+C.

Question 15

The product of the positive integers 1 to 4 is: $4 \times 3 \times 2 \times 1 = 24$

and can be written in an abbreviated form as 4!. We say "4 factorial". So 4! = 24.

The product of the positive integers 1 to 16 is: $16 \times 15 \times 14 \times .... \times 3 \times 2 \times 1$and can be written in an abbreviated for as 16! We say "16 factorial". The ... represents the product of all the missing integers between 14 and 3. In general, the product of the positive integers 1 to n is n!. Note that 1!= 1.

Determine the last two digits of the sum:

1! + 2! +3! +...+2019! + 2020! + 2021!

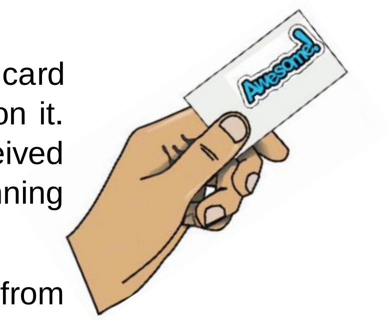
Question 16

There are 90 000 five-digit positive integers. However, only some of these five-digit integers satisfy the following conditions:

- the middle digit is 0,
- the ten thousands digit and the tens digit are equal,
- the thousands digit and the ones (units) digit are equal, and
- the number has exactly 5 prime factors. All of these prime factors are odd and none are repeated.

Determine the smallest five-digit positive integer that satisfy the given conditions

Question 17



Every student in a class at Sunshine High School committed to sending a card to every other student in the class with some word of encouragement on it. The students that have a first name beginning with a letter from A to K received a total of 459 cards. The remaining students that have a first name beginning with a letter from L to Z received a total of 297 cards.

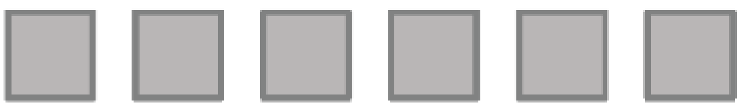
How many students in the class have a first name beginning with a letter from A to K?

Question 18

A six-digit code is required to gain access to a secure building. You know the following about the correct code:

- only the digits 1, 2, 3, 4, 5, and 6 are used in the correct code and each of these digits appears exactly once;
- the fifth digit in the code is not a 5; the sixth digit in the code is not a 6; and
- there is only one correct code.

What is the probability that you are able to enter the correct code?



Question 19

A jar contains only small red and small yellow candies. Another 30 red candies are added to the candies already in the jar so that one-third of the total number of candies in the jar are red candies.

At this point, 30 yellow candies are added to the jar, and now three-tenths of the total number of candies in the jar are red candies. What fraction of the number of the candies originally in the jar were red candies?

Question 20

A spiral of numbers is created, as shown, starting with 1. If the pattern of the spiral continues, how will the numbers 2020, 2021, and 2022 appear in the spiral?

$$\begin{array}{ccccccccc}
 & & 10 & \to & 11 & \to & 12 & \to & 13\\
 & & \uparrow & & & & & & \downarrow\\
 & & 9 & & 2 & \to & 3 & & 14\\
 & & \uparrow & & \uparrow & & \downarrow & & \downarrow\\
\vdots & & 8 & & 1 & & 4 & & 15\\
\uparrow & & \uparrow & & & & \downarrow & & \downarrow\\
22 & & 7 & \leftarrow & 6 & \leftarrow & 5 & & 16\\
\uparrow & & & & & & & & \downarrow\\
21 & \leftarrow & 20 & \leftarrow & 19 & \leftarrow & 18 & \leftarrow & 17
\end{array}$$

A. Left to right in a row.
B. Right to left in a row.
C. Down in a column.
D. Up in a column.
E. Down and then left.
F. Up and then right.